

Enhancing RAG Pipeline Performance with Translation-Based Embedding Strategies for Non-English Documents

Can Iscan*, Muhammet Furkan Ozara*, Akhan Akbulut*[†]

*R&D Center of Next4biz Sahrayicedit Mah, Pakpen Plaza, No:40/4, 34734 Istanbul, Turkey
can.iscan@next4biz.com, furkan.ozara@next4biz.com, akhan.akbulut@next4biz.com

[†]Department of Computer Engineering Istanbul Kültür University 34536 Istanbul, Turkey
a.akbulut@iku.edu.tr

Abstract—Despite the advances in multilingual embedding models, they often underperform compared to embedding documents in English, which limits the effectiveness of Retrieval-Augmented Generation (RAG) systems in non-English question-answering contexts. This disparity poses a significant barrier for non-English speakers to access high-quality AI-driven retrieval systems. To address this challenge, this paper proposes a novel approach that leverages English embeddings enhanced by translation. Our approach involves translating texts into English and then embedding them while preserving the original information as metadata. By utilizing robust English and multilingual embedding models, we achieve superior results in RAG systems. We present a comprehensive approach for constructing a question-answering system using GPT-4o technology, which leverages translation models to enhance English embeddings. We conducted actual studies to compare the suggested methodology with conventional methods using various embedding models, employing the RAGAS framework. Initial findings indicate that our method greatly improves the retrieval effectiveness of RAG systems in non-English environments, attaining higher context precision and recall metrics. This adaptable solution incorporates English embeddings into multilingual applications, providing a versatile and effective method to improve non-English scenarios by using the capabilities of translation models.

Index Terms—RAG, question-answering, multilingual embeddings, translation, ChatGPT, GPT-4o

I. INTRODUCTION

Today's world is getting revolutionized by the advancements made in artificial intelligence, machine learning, generative adversarial networks, and natural language processing [1]. In the past, no one would believe that a mathematical formula inside some chips could compose music, draw glamorous artworks, generate photorealistic video clips, write a hundred pages long novels, solve the hardest math problems, and drive cars on highways [2]. Among all of those capabilities that the current state of machine learning models can do, the most fascinating and may be the most helpful one is chatbots [3].

The popularity of chatbots rised at the end of 2022 with the release of ChatGPT. This product showed people how far artificial intelligence has come by its knowledge of nearly the whole internet, the intelligence of understanding questions and

with its replies, also how practical it is [4]. The underlying technology powering these chatbots is comprised of sophisticated language models, which are trained using vast amounts of textual data sourced from various internet platforms such as blogs, encyclopedias, e-books, and forums.

Chatbot systems are being more commonly included into different platforms including Enterprise Resource Planning (ERP), Customer Relationship Management (CRM), and Customer Service Management (CSM) solutions to offer effective and scalable customer support. These solutions are incredibly practical since they aid clients in meeting their needs, enhancing overall satisfaction and efficiency. We employ data from the Next4Biz CSM platform to assess our methodology in this study. The evaluation is carried out using human-generated questions that are exclusively focused on Next4Biz CSM, guaranteeing the pertinence and precision of our assessment.

Large language models (LLMs) and chatbots exhibit high performance when provided with training data that covers a substantial percentage of the internet. Nevertheless, adopting new information into LLMs continues to be a difficult task, despite the utilization of methods such as fine-tuning [5]. For example, a chatbot that is included into a company's customer care platform may lack specialized information regarding the company's products. One potential solution to this issue is the implementation of Retrieval Augmented Generation (RAG) pipelines [6].

RAG pipelines facilitate the acquisition of new knowledge by chatbots in particular areas by finding the most pertinent documents and integrating them into the input for the LLM. This method has enabled the invention of sophisticated question-answering systems, bots for customer service, and tools for content production and summarization. The embedding models are a fundamental element of RAG systems as they are responsible for extracting the semantic meaning of the input text. While embedding models are created for both pure English and multilingual domains, they tend to exhibit superior performance in English, even when the model is designed to handle many languages [7]. Utilizing powerful English embedding models or attaining the same level of performance as English embeddings in multilingual models

for other languages can improve the performance of RAG applications.

This research introduces a new approach to enhance the effectiveness of English embeddings in multilingual models. It also explores the utilization of English embedding models for non-English documents to enhance the performance of RAG in question-answering domains. We utilize multiple English and multilingual embedding models in a question-answering pipeline powered by GPT-4. We evaluate their performance using the RAGAS framework. Our methodology consists of employing translation models to convert texts into English while preserving the original language version as metadata. We then do retrieval using the English-translated versions and utilize the untranslated version within the appropriate context.

The primary contributions of this research are four-fold:

- We demonstrate the potential of English embedding models to improve RAG performance in multilingual scenarios by using translation.
- By translating non-English documents into English and embedding them with multilingual embedding models, we demonstrate the performance advantages over traditional methods.
- We benchmark multiple English and multilingual embedding models using the suggested approach to find the most efficient and accurate RAG pipeline models in non-English question-answering systems and compare their performance.
- We provide a comprehensive analysis of the cost-benefit ratio of implementing the proposed RAG pipeline, offering valuable guidance for practical applications and future research.

The paper is organized as follows. In Section II, we discuss the related research work in the RAG field and highlight similar approaches to achieve better retrieval performance. Section III explains the proposed model and the techniques that are employed. In Section IV, we analyze the results of the experiments, comparing the performance of different embedding models and discussing the impact of using English embeddings on non-English question-answering tasks. Section V presents an in-depth analysis of the findings, and possible constraints and discusses the applicability and scalability of the solution. In Section VI, we summarize the key points of this research and give future research directions.

II. RELATED WORK

Researchers and AI engineers from industry have made tremendous progress in the topic of RAG, researching various use cases to improve the capabilities of LLMs. RAG systems enhance the precision of LLM-generated answers and also facilitate the use of LLMs in new or specific fields without requiring lengthy retraining. RAG systems surpass the constraints of LLMs that are exclusively trained on existing data by including supplementary documents like QA text pairings or complete books. The vast range of applications, such as customer care bots, complex question-answering systems, and

content production tools, has resulted in the widespread use of RAG.

Researchers and AI engineers from the industry have investigated many applications of RAG in the literature. In addition to enhancing the precision of responses given by LLMs, RAG allows LLMs to be utilized for novel or specific applications without requiring additional training. RAG systems enhance the capabilities of LLMs by including other documents such as QA text pairs or complete books. This expansion enables them to produce text that is more contextually appropriate and precise, surpassing the limitations of their training data. RAG systems have broad applications in various domains, including customer support bots, sophisticated question-answering systems, and content production tools.

Prior to the introduction of RAG approaches in the literature, major improvements in the field of natural language processing (NLP) [8], [9] and embedding models established the foundation for the advanced RAG systems that are currently employed. Mikolov et al. were the first to develop embedding models, which convert texts into low-dimensional vectors that capture semantic meaning. They introduced the 'word2vec' model as a way to learn word representations [10]. Later developments included the introduction of models such as GloVe [11] and fastText [12]. Although the majority of these models were first trained using English data, subsequent development led to the creation of multilingual embedding models that can function across many languages. Nevertheless, these multilingual models generally exhibit worse performance in comparison to models purely trained in English [13].

Several approaches have been suggested to get better performance from RAG on multilingual documents. The most common way is to use a multilingual embedding model, but these models are generally considered subpar compared to their English-only counterparts [14]. Another study introduced a framework for multilingual keyphrase generation with iterative retriever-generator training, further illustrating the potential of leveraging advanced techniques in multilingual NLP tasks [15].

Translation-based RAG methods involve translating the documents from the target language to English and can utilize the English embedding models. Again, this demands very high-quality translation models as the goodness of the translations affects directly on the quality of the overall performance [16]. Recent studies now suggest that large-scale multilingual NMT models, trained using large amounts of parallel data across a large number of languages, can indeed raise translation quality for both high- and low-resource languages. For instance, Google has recently conducted work on massively multilingual NMT; it found that considerable improvements in quality arise when models are trained on a large, highly multilingual corpus [17].

Researchers have investigated human-like translation strategies using LLMs, which have shown to improve translation quality. Techniques such as Multi-Aspect Prompting and Selection (MAPS) replicate the nuanced processes of human translators, offering more accurate translations by utilizing

context and minimizing issues like hallucinations and ambiguities. These advancements highlight the necessity of high-quality translation models and sophisticated methods for effective RAG systems [18].

In our opinion, RAG has greatly improved LLMs, allowing them to be used in many fields without lengthy retraining. The cornerstone for these advances was early embedding models like 'word2vec,' GloVe, and fastText, which were trained on English data. Mixed-language models generally underperform English-only models. Innovative translation tactics and translation-based methodologies strive to close this gap, showing RAG systems' potential for customer support and content production.

III. METHODOLOGY

To investigate the effectiveness of using English embeddings for non-English documents in RAG pipelines, we designed a comprehensive experimental setup involving real-world documents, a vector store, a translation mechanism, an evaluation framework, and multiple embedding models.

A. Documents

This research involved an in-depth analysis of 88 Turkish documents sourced from the Next4biz CSM platform user manual, which encompasses detailed instructions and descriptions for various software products. The primary objective was to enhance the question-answering system's ability to deliver precise and contextually relevant responses to user inquiries regarding the Next4biz CSM software. We developed a RAG pipeline to empower the LLM to effectively access, retrieve, and extract pertinent information from the extensive content within these documents. By integrating this RAG pipeline, the system is capable of generating accurate and comprehensive answers, ensuring that users receive the most relevant and reliable information in response to their queries.

B. Vector store

Vector stores are one of the key components of the RAG systems. Vector stores, or vector databases, provide an efficient way to store and retrieve high-dimensional vectors. In general, vector stores are used with an embedding model to allow storing and retrieving processes more convenient. These vector embeddings represent text, images, or other data types in a dense format that captures semantic meaning, enabling rapid similarity searches. By leveraging vector stores, RAG systems can efficiently handle large volumes of data, quickly find relevant information, and provide more accurate and relevant responses. In this study, we took advantage of one of the most popular vector stores, ChromaDB [19].

ChromaDB is an open-source vector store used to store and retrieve vector embeddings. It is highly optimized for performance and scalability, making it the optimal choice for RAG applications. Additionally, It can be seamlessly integrated with popular machine learning frameworks and embedding models, providing flexibility in generation and retrieval steps. In our study, leveraging ChromaDB allows us to fairly benchmark the proposed approach with the chosen embedding models.

C. Embedding Models

Embedding models are machine learning models which compress the high-dimension data such as texts and images, into lower-dimension vectors. These produced vectors capture the semantic meaning of the input, which allows to representation of similar concepts with non-identical inputs. Embedding models are highly used in natural language tasks such as translation, classification, summarization, and relevant text retrieval.

Embedding models work by learning a mapping function that reduces high-dimensional data into a lower-dimensional space. Embedding models aim to comprehend the meanings of various terms in context, attempting to discern the meanings of different words when combined. Examining the frequency of two words appearing together within a corpus of texts enables this process. Words that co-occur often tend to have related meanings or synonyms. Using such ideas, the embedding matrix would place these terms close together in Euclidean space. In the context of question-answering applications, embedding models are utilized to embed documents that contain the necessary information for answering and also embed the question to find the closest documents to the asked question. Afterwards, those pertinent documents are passed to LLM with the question so that the answer can be correctly generated using the selected documents. By locating an embedding model within a vector store, the documents can be easily embedded and stored, and the most relevant documents can be efficiently retrieved during question-answering. Embedding models are categorized based on the languages they are designed to process. Multilingual embedding models are trained on large volumes of text, including various languages, which allows them to represent words and concepts across different languages in a single embedding space. However, multilingual models might struggle to capture some of the meaning from the text compared to English embedding models, which are trained with corpus containing only English. The choice between these types of embedding models is made depending on the application and language requirements.

In our research, we investigate the performance gain of utilizing English embeddings in a question-answering pipeline for non-English languages, such as Turkish, through the application of English and multilingual embedding models. For this purpose, we included several multilingual and English models in the experimental setup. Large Text Embedding Benchmark (MTEB) [20] testing results for the tested embedding models are shown in Table I.

D. Proposed Approach

RAG pipelines used in question-answering systems generally follow similar architectures to each other. A traditional RAG pipeline has 4 main roles, which are user, vector store, embedding model, and LLM. The RAG pipeline's flow begins with receiving a question from the user. This question is passed to a retrieval system, which is made from a vector store, to run a similarity search and retrieve the most relevant texts to the question. The system then passes the retrieved paragraphs and

TABLE I
EVALUATED EMBEDDING MODELS FOR MULTILINGUAL AND ENGLISH
TEXT PROCESSING

Model Name	Language	Model Size (Million Params)	Embedding Space	Retrieval Score (English)
OpenAI/ text-embedding-3-large	Multilingual	-	3072	55.44
OpenAI/ text-embedding-3-small	Multilingual	-	1536	51.08
Alibaba-NLP/ gte-large-en-v1.5	English	434	1024	57.91
intfloat/ multilingual-e5-large	Multilingual	560	1024	51.43
Snowflake/ snowflake-arctic-embed-l	English	334	1024	55.98

the question to a Learning Machine Model (LLM) to generate an answer. Finally, the system sends the generated answer from an LLM back to the user as a response. This flow is depicted in Figure 1.

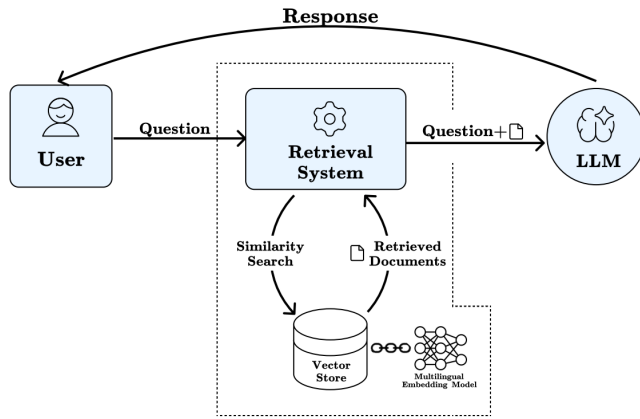


Fig. 1. Traditional RAG pipeline

The traditional approach requires the use of a multilingual embedding model to handle non-English documents. Multilingual models perform relatively poorly compared to pure English models. Also, some of the multilingual models achieve better retrieval performances in English compared to non-English languages. These issues necessitate the incorporation of non-English documents, written in English, into both English and multilingual models. We solved this problem by placing a translator in front of vector space.

Our proposed approach harnesses the power of state-of-the-art translation models to enable the utilization of English embeddings for non-English documents. Before letting The selected translation model must populate the vector store with translated English documents before allowing the pipeline to ansOur proposed model's question-answering flow starts by retrieving a question from the user and passing it to the retrieval system. The selected translation model then translates the question to English. We conducted the similarity search by embedding the translated question and choosing the closest documents. The metadata attachments provide the original versions of the retrieved documents. We package

the question and the retrieved documents together before sending them to LLM for answer generation. This flow of the proposed approach is presented in Figure 2

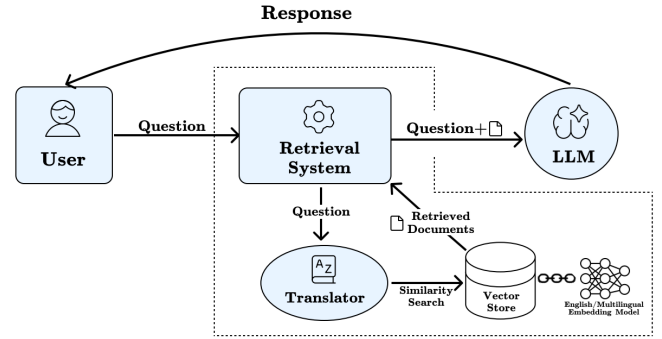


Fig. 2. Translator adapted RAG pipeline

IV. EXPERIMENTAL RESULTS

In this section, we present the results of our experiments. The RAG of the proposed approach as opposed to the traditional RAG approach for non-English use cases has been evaluated with the RAG Assessment (RAGAS) framework [21]. RAGAS is a popular benchmarking framework for evaluating RAG pipelines. To evaluate the effectiveness of our approach, we have integrated RAGAS with the RAG pipeline we developed specifically for this research. RAGAS is utilizing LLM models to assess the performance of question-answering bots. This is done by posing questions related to the recovered context and the original question, and determining if the retrieved context is adequate to answer the question. In order to enhance response generation, we have integrated RAGAS with GPT-4o, since we also utilize that LLM model. Utilizing a diminutive LLM model in RAGAS while employing a more substantial model in the RAG pipeline leads to deceptive metric outcomes. We have primarily concentrated on two factors concerning retrieval performance, which are explained below.

a) *Context Recall*: Context recall is a measure of how well the retrieved context aligns with the ground truth answer. The metric score falls between 0 and 1, where higher values indicate better performance. To calculate context recall, each sentence in the ground truth answer is analyzed to determine if it can be referenced from the retrieved context. Ideally, the retrieved context should encompass all sentences in the ground truth answer.

b) *Context Precision*: Context precision assesses the ranking of all ground-truth relevant items in the contexts. We compute it using the question, ground truth, and the retrieved contexts. The metric also ranges between 0 and 1, where higher scores indicate better precision. In an ideal scenario, all relevant chunks should appear at the top ranks of the retrieved contexts.

We begin the experimental process by collecting Turkish documents from the Next4biz CSM products manual

and parsing them. These documents are then translated into English using a state-of-the-art neural machine translation model, *Helsinki-NLP/opus-mt-tc-big-en-tr* from the OPUS-MT project [16]. English embedding models then process the translated documents to generate semantic embeddings. We employ a chroma vector database to store the embeddings of the documents. The original Turkish version of the documents is retained as metadata to be included in the final retrieval step.

We evaluated the proposed approach and the traditional approach for retrieving non-English documents, using different embedding models to reduce the bias arising from an embedding model. Multilingual test multilingual embedding models with both approaches, while we only test English models with the proposed approach. n-written questions and related answers are prepared and used with the RAGAS to obtain a measurable result. RAGAS used the RAG pipeline to generate answers for the questions and leveraged GPT-4o to evaluate if the return context is adequate to answer the question. The experiments on the proposed approach revealed that the embedding models obtained the same context recall score of 0.76. The best-performing model among the selected ones was OpenAI’s text-embedding-large-3, which had a context recall score of 0.76 and context precision of 0.65. The context precision metric represents the performance difference among the tested embedding models in the ordering of the returned documents. In Table II, the experimental results of the traditional approach are shown.

TABLE II
PERFORMANCE EVALUATION OF EMBEDDING MODELS USING
TRADITIONAL APPROACH

<i>Model Name</i>	<i>Language</i>	<i>Context Recall</i>	<i>Context Precision</i>
intfloat/ multilingual-e5-large	Multilingual	0.76	0.64
OpenAI/ text-embedding-3-large	Multilingual	0.76	0.65
OpenAI/ text-embedding-3-small	Multilingual	0.76	0.61

According to the evaluation results of the proposed approach, OpenAI’s text-embedding-3-large model acquired the best metric scores of 0.81 in context recall and 0.72 in context precision. The second best-performing model is found to be *gte-large-en-v1.5*, which closely follows the champion model in metric scores and is also much smaller in embedding dimension size. The embedding model from Snowflake obtained the lowest metric scores, but it is also the smallest model in parameter size among the selected models. The proposed approach’s results are presented in Table III.

When both approaches are compared with the results of the assessment, OpenAI’s *text-embedding-3-large* model stands out with the best context recall and precision scores of 0.81 and 0.72. The proposed approach outperformed the traditional approach in the text-embedding-3-large model by approximately 10%. However, when intfloat’s *multilingual-e3-large*

TABLE III
PERFORMANCE EVALUATION OF EMBEDDING MODELS USING PROPOSED
APPROACH

<i>Model Name</i>	<i>Language</i>	<i>Context Recall</i>	<i>Context Precision</i>
intfloat/ multilingual-e5-large	Multilingual	0.76	0.58
Alibaba-NLP/ gte-large-en-v1.5	English	0.80	0.69
Snowflake/ snowflake-arctic-embed-l	English	0.44	0.28
OpenAI/ text-embedding-3-large	Multilingual	0.81	0.72
OpenAI/ text-embedding-3-small	Multilingual	0.76	0.71

model is employed within the proposed approach, a drop in context precision is observed. The English embedding model *gte-large-en-v1.5* achieved better metric scores compared to *multilingual-e5-large*, even though it has fewer parameters than the other. Additionally, it achieved a better context recall score than the *text-embedding-3-small* model.

V. DISCUSSION

In the current state of question-answering pipelines, retrieval-augmented generation is one of the most used techniques to feed LLM models with new information. The most crucial part of the RAG pipelines is embedding models used while storing and retrieving the relevant documents. Employing larger and more advanced embedding models improves the retrieval of more similar documents to the question, leading to more accurate responses. In this research work, we evaluated several different embedding models using both the proposed and traditional RAG approaches to determine whether translating non-English documents enhances retrieval performance. The assessment presents that the multilingual embedding models, such as OpenAI’s text-embedding-3-large, do not perform as well with Turkish documents compared to English documents. Also, the smaller English embedding model accomplished better results compared to other multilingual models with bigger parameter sizes. The findings revealed that the translation of non-English documents to English has a positive impact on retrieval performance when utilizing either a multilingual or English embedding model.

The results of the experiments show that multilingual models perform better in the English language compared to other or less-used languages, such as Turkish. The retrieval scores obtained in the experiments have revealed that the information loss occurring in the translation of the documents is not higher than the performance gain acquired using the proposed approach. The utilization of better translation models/techniques could decrease the information loss, allowing the proposed approach to perform even better in retrieval. The carried out experiments underscore the importance of embedding model selection. The experiments revealed significant performance differences among English and multilingual embedding mod-

els themselves. This highlights the need for a careful selection process tailored to the specific requirements of the application. In this study, models with a smaller number of parameters that were optimized well for specific tasks proved more effective than larger generalized ones.

We constructed the experiments in this study using Turkish documents, which yielded promising results when we applied the proposed approach. Testing the approach with other languages can yield varying results and best-performing models. However, for less-used languages, the proposed approach outperforms the traditional approach. In this case, the data used for the embedding model's training is the most decisive factor. If the data contains a large amount of the used language, the performance difference between it and English tends to be smaller. The response times between traditional and proposed approaches differ slightly because a translation model is located in front of the vector store. In real time, the translation model needs to translate the incoming questions into English. The metadata stores the original versions of the retrieved documents from the vector store, eliminating the need for translation. The response time difference between approaches is only due to the translation of the questions, which are typically a sentence long. Translation doesn't pose a speed issue unless it involves the use of larger models.

VI. CONCLUSION

In this work, we developed an innovative strategy to enhance RAG performance by shifting retrieval processes to English for non-English documents through the application of translation models. We evaluated this strategy using multiple embedding models and the RAGAS framework, testing it with 15 question-answer pairs. Our findings show that this approach significantly outperforms traditional methods in terms of context precision and recall. The OpenAI text-embedding-3-large multilingual model achieved a context recall of 0.81 and a context precision of 0.73 with our method, marking a 10% improvement over the traditional method's context recall of 0.76 and context precision of 0.65. This advancement underscores the effectiveness of leveraging high-quality English embedding models for non-English tasks and suggests that further enhancements could be made with more advanced translation models. Our inspiration came from approaches that leverage deep learning methods for software vulnerability prediction [22], which have proven to offer competitive alternatives to existing techniques in other domains.

REFERENCES

- [1] Immad A Shah and SukhDev Mishra. Artificial intelligence in advancing occupational health and safety: an encapsulation of developments. *Journal of Occupational Health*, 66(1):uiad017, 01 2024.
- [2] Sorin Grigorescu, Bogdan Trasnea, Tiberiu Cocias, and Gigel Macesanu. A survey of deep learning techniques for autonomous driving. *Journal of Field Robotics*, 37(3):362–386, 2020.
- [3] Patrick S. H. Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. *CoRR*, abs/2005.11401, 2020.
- [4] Christoph Leiter, Ran Zhang, Yanran Chen, Jonas Belouadi, Daniil Larionov, Vivian Fresen, and Steffen Eger. Chatgpt: A meta-analysis after 2.5 months, 2023.
- [5] Andy Wang, Cong Liu, Jingye Yang, and Chunhua Weng. Fine-tuning large language models for rare disease concept normalization. *Journal of the American Medical Informatics Association*, page ocae133, 06 2024.
- [6] Huayang Li, Yixuan Su, Deng Cai, Yan Wang, and Lemao Liu. A survey on retrieval-augmented text generation. *CoRR*, abs/2202.01110, 2022.
- [7] Chihiro Yano, Akihiko Fukuchi, Shoko Fukasawa, Hideyuki Tachibana, and Yotaro Watanabe. Multilingual sentence-t5: Scalable sentence encoders for multilingual applications, 2024.
- [8] Sera Deniz Sosun, Bulent Tayfun, Yasemin Nukan, İrem Altun, Elif Berra Erik, Elif Yldrm, Büşra Kocaçınar, and Fatma Patlar Akbulut. Deep sentiment analysis with data augmentation in distance education during the pandemic. In *2022 Innovations in Intelligent Systems and Applications Conference (ASYU)*, pages 1–5. IEEE, 2022.
- [9] Büşra Kocaçınar, Nasibullah Qarizada, Cihan Dikkaya, Emirhan Azgun, Elif Yıldırım, and Fatma Patlar Akbulut. Analysis of the lingering effects of covid-19 on distance education. In *IFIP International Conference on Artificial Intelligence Applications and Innovations*, pages 189–200. Springer, 2023.
- [10] Tomas Mikolov, Kai Chen, Greg Corrado, and Jeffrey Dean. Efficient estimation of word representations in vector space, January 2013.
- [11] Jeffrey Pennington, Richard Socher, and Christopher Manning. GloVe: Global vectors for word representation. In Alessandro Moschitti, Bo Pang, and Walter Daelemans, editors, *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 1532–1543, Doha, Qatar, October 2014. Association for Computational Linguistics.
- [12] Piotr Bojanowski, Edouard Grave, Armand Joulin, and Tomas Mikolov. Enriching word vectors with subword information. *Transactions of the Association for Computational Linguistics*, 5:135–146, 2017.
- [13] Daniel Cer, Mona Diab, Eneko Agirre, Iñigo Lopez-Gazpio, and Lucia Specia. SemEval-2017 task 1: Semantic textual similarity multilingual and crosslingual focused evaluation. In Steven Bethard, Marine Carpuat, Marianna Apidianaki, Saif M. Mohammad, Daniel Cer, and David Jurgens, editors, *Proceedings of the 11th International Workshop on Semantic Evaluation (SemEval-2017)*, pages 1–14, Vancouver, Canada, August 2017. Association for Computational Linguistics.
- [14] Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. Unsupervised cross-lingual representation learning at scale. *CoRR*, abs/1911.02116, 2019.
- [15] Yifan Gao, Qingyu Yin, Zheng Li, Rui Meng, Tong Zhao, Bing Yin, Irwin King, and Michael R. Lyu. Retrieval-augmented multilingual keyphrase generation with retriever-generator iterative training, 2022.
- [16] Jörg Tiedemann and Santhosh Thottingal. OPUS-MT – building open translation services for the world. In André Martins, Helena Moniz, Sara Fumega, Bruno Martins, Fernando Batista, Luisa Coheur, Carla Parra, Isabel Trancoso, Marco Turchi, Arianna Bisazza, Joss Moorkens, Ana Guerberof, Mary Nurminen, Lena Marg, and Mikel L. Forcada, editors, *Proceedings of the 22nd Annual Conference of the European Association for Machine Translation*, pages 479–480, Lisboa, Portugal, November 2020. European Association for Machine Translation.
- [17] Sneha Reddy Kudugunta, Ankur Bapna, Isaac Caswell, Naveen Ariavazhagan, and Orhan Firat. Investigating multilingual NMT representations at scale. *CoRR*, abs/1909.02197, 2019.
- [18] Zhiwei He, Tian Liang, Wenxiang Jiao, Zhuosheng Zhang, Yujiu Yang, Rui Wang, Zhaopeng Tu, Shuming Shi, and Xing Wang. Exploring Human-Like Translation Strategy with Large Language Models. *Transactions of the Association for Computational Linguistics*, 12:229–246, 03 2024.
- [19] Chroma Team. Chroma documentation. Online, 2024. docs.trychroma.com.
- [20] Niklas Muennighoff, Nouamane Tazi, Loïc Magne, and Nils Reimers. Mteb: Massive text embedding benchmark, 2023.
- [21] Shahul Es, Jithin James, Luis Espinosa-Anke, and Steven Schockaert. Ragas: Automated evaluation of retrieval augmented generation, 2023.
- [22] Çagatay Catal, Akhan Akbulut, Sašo Karakatič, Miha Pavlinek, and Vili Podgorelec. Can we predict software vulnerability with deep neural network? In *Proceedings of the 19th International Multiconference on Information Society, Ljubljana, Slovenia*, pages 9–13, 2016.